

Genomic analysis leadership for therapeutic discovery

Syed Abbas Bukhari, PhD | Industry and research institute research plan

My contribution is to lead genomic analyses that turn complex datasets into testable therapeutic hypotheses and useful research decisions. My career connects comparative neurogenomics, single-cell biology, and pediatric oncology through a consistent question: how do regulatory programs link molecular variation to phenotype? I would align this expertise with the organization's disease priorities, available data, and experimental capabilities.

Evidence of readiness

At NCI, I lead computational analysis of a rhabdomyosarcoma and muscle-reference atlas, from expression-program discovery through regulatory interpretation and projection into clinically annotated cohorts. I also contribute computational leadership to a pan-pediatric project combining regulatory networks, functional dependencies, and pharmacologic evidence. Both manuscripts are in preparation. These projects demonstrate analytical ownership, reproducible implementation, and biological interpretation. Current outcome associations remain exploratory, not validated prognostic biomarkers.

Build reliable disease and cohort analyses

I would define the decision an analysis must support with program scientists, then assess sample provenance, clinical annotations, assay quality, and confounding. Integrating single-cell and bulk expression with available genomic or chromatin evidence would distinguish recurrent disease programs from composition, proliferation, and copy-number effects. Deliverables would include harmonized analysis-ready datasets, annotated cell-state references, documented quality control, and reports identifying reproducible findings and the additional evidence needed. These outputs would help teams select informative cohorts and experimental models.

Prioritize targets and mechanisms for experimental follow-up

I would combine transcription-factor and regulatory-network evidence with functional dependencies and compound information to rank candidate vulnerabilities against the program's therapeutic objectives. Ranking would consider disease specificity, generic essentiality, model relevance, contradictory evidence, and uncertainty. With experimental investigators, I would translate the strongest hypotheses into perturbation comparisons and expected readouts. Deliverables would be traceable candidate evidence summaries and analysis plans supporting decisions to advance a target, refine a mechanism, or stop pursuing a weak hypothesis. Computational nomination would not be presented as target validation.

Evaluate response hypotheses and deliver reusable analyses

Where suitable treatment and outcome data exist, I would test molecular correlates of response or resistance using prespecified comparisons, clinical covariates, appropriate diagnostics, and independent validation when feasible. Candidate biomarker assessments would distinguish association from predictive performance and treatment benefit. I would lead version-controlled R/Python workflows, reproducible computing environments, reviewable reports, and handoff documentation, while mentoring colleagues through analysis planning and code review. Milestones would be agreed with program leads and adjusted to data readiness and experimental results. All work would use employer-authorized data and access arrangements.

Success would mean reliable genomic analyses, clearer experimental priorities, and reusable computational capabilities that help the organization advance promising therapeutic hypotheses and avoid investing in poorly supported ones.